THIRD-PARTY DATA PIPELINE STRATEGY
Horizon Tech — VRU Trading Bot Research
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CORE THESIS

The data gap is the primary obstacle between VRU and institutional-grade performance. Piping in third-party alternative data directly into the training and inference pipeline closes this gap without requiring decades of proprietary labeled history.

DATA TIERS

TIER 1 — ESSENTIAL (highest alpha signal)

Options Flow
  Sources: Unusual Whales, Market Chameleon
  Why: Institutions hedge BEFORE they move. Large unusual call sweeps precede accumulation by hours to days. This is the earliest detectable signal.

Dark Pool Prints
  Sources: Quiver Quant, Blackbox Stocks
  Why: Institutional block trades that bypass lit exchanges still appear in consolidated tape reporting. These prints reveal positioning before price moves.

Level 2 / Order Book Imbalance
  Why: Bid-ask pressure asymmetry is the most direct, real-time institutional footprint. Persistent imbalance on one side without price movement = accumulation signal.

TIER 2 — HIGH VALUE

13F Filings Delta
  Sources: SEC EDGAR, Quiver Quant
  Why: Quarterly, so not real-time — but useful for confirming patterns the model learned from Tier 1 signals. Ground truth validation for training labels.

Short Interest Flow
  Why: Sudden drops in short interest signal institutional covering.
  Sustained high short interest with price stability = institutional support beneath.

ETF Creation/Redemption Baskets
  Why: BlackRock's own ETF mechanics leak positioning. When authorized participants create or redeem large ETF baskets, underlying constituent demand is visible.

TIER 3 — EDGE (supplementary)

Congressional Trading Disclosures
  Sources: Quiver Quant, Capitol Trades
  Why: Not institutional, but follows similar informed-buyer footprint logic.
  Legally required disclosure creates a lagged but exploitable signal.

Earnings Call Sentiment Velocity
  Why: Institutions front-run guidance changes. NLP on call transcripts with
  velocity weighting (how fast sentiment shifts) precedes positioning.

PIPELINE ARCHITECTURE

Raw Data → Normalization → Feature Engineering → VRU Multi-Task Core → Signal Output

Normalization strategy:
  - All price/volume inputs: percent change, not absolute values (stationarity)
  - Options flow: normalize by 30-day average volume for that ticker
  - Dark pool prints: normalize by float size
  - Order imbalance: scale to [-1, 1] per rolling window

Feature engineering:
  - Rolling Z-scores for anomaly detection on each stream
  - Lag features: t-1, t-5, t-10 for each input
  - Cross-stream correlation features (options + dark pool co-occurrence)

Output head:
  - Multi-class classifier: Flat / Accumulation / Distribution
  - Confidence score per class (softmax probabilities)
  - Confidence score feeds position sizing

INTEGRATION NOTE

Unusual Whales, Quiver Quant, and Blackbox Stocks all have REST APIs.
Direct pipeline into VRU inference requires:
  1. Real-time websocket or polling connection per data source
  2. Shared normalization layer across all streams
  3. Feature vector assembly before each forward pass
  4. Signal output → execution layer (broker API)

The VRU's multi-stream telemetry architecture from v23 is the correct template.

The 3-stream (power + thermal + network) experiment maps directly to (price + options flow + dark pool) with the same lag coupling structure.